TVA TENNESSEE VALLEY AUTHORITY

Post Office Box 2000, Decatur, Alabama 35609-2000

October 12, 2023

10 CFR 50.73
10 CFR 50.4(a)

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Unit 1
Renewed Facility Operating License No. DPR-33
NRC Docket No. 50-259

Subject: **Licensee Event Report 50-259/2022-002-01 – High Pressure Coolant Injection System Declared Inoperable Due to a Corroded Actuator**

Reference:
1. Non-Emergency Event Notification 55992 – High Pressure Coolant Injection Inoperable
2. TVA letter to NRC, "Licensee Event Report 50-259/2022-002-00 – High Pressure Coolant Injection System Declared Inoperable Due to a Corroded Actuator," dated September 12, 2022 (ML22255A227)

In the reference letter, Tennessee Valley Authority (TVA) submitted the subject Licensee Event Report (LER), which provided details of the inoperability of the Browns Ferry Nuclear Plant (BFN), Unit 1, High Pressure Coolant Injection (HPCI) System due to a corroded actuator. TVA submitted this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(v)(D), as any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. As noted in the referenced letter, TVA planned to supplement this LER to provide additional time to complete the root cause evaluation.

Accordingly, enclosed is a supplement to the subject LER (i.e., LER 50-259/2022-002-01), which provides additional detail into the causal factors of the event and includes additional corrective actions to prevent recurrence.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact David Renn, Site Licensing Manager, at (256) 729-2636.

Respectfully,

Manu Sivaraman
Site Vice President

Enclosure: Licensee Event Report 50-259/2022-002-01 – High Pressure Coolant Injection System Declared Inoperable Due to a Corroded Actuator

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

NRC FORM 366 (08-2020)

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023

1. Facility Name		2. Docket Number	3. Page
Browns Ferry Nuclear Plant, Unit 1	☒ 050 ☐ 052	00259	1 OF 8

4. Title
High Pressure Coolant Injection System Declared Inoperable Due to a Corroded Actuator

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
07	12	2022	2022	- 002 -	01	10	05	2023	N/A	05000 N/A
									N/A	05000 N/A

9. Operating Mode	10. Power Level
1	097

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Ryan Coons, Licensing Engineer	256-729-2070

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
D	BJ	FCV	T147	Y	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date		N/A	N/A	N/A

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On July 12, 2022, during a routine surveillance test, the Browns Ferry Nuclear Plant (BFN), Unit 1, high pressure coolant injection (HPCI) turbine control (governor) valve did not open as expected. Upon investigation, it was discovered that the electro-governor remote hydraulic actuator (EGR) had internal corrosion, which resulted in binding of the pilot plunger and associated pilot bushing and a subsequent failure to supply the hydraulic oil needed to open the governor valve.

The resulting inoperability of the single-train HPCI system would have prevented HPCI from fulfilling its safety function to mitigate the consequences of an accident if demanded.

The event was caused by inadequate procedural guidance which led to failing to identify corrosion which led to the valve failure. Additionally, BFN's maintenance strategy was not aligned with industry best practices and was insufficient for making risk-based decisions related to the reliability of the U1 HPCI EGR. The EGR and remote servo were promptly replaced. As corrective actions to prevent recurrence, BFN revised maintenance procedures to include enhanced acceptance criteria (both wording and images) and now requires photographing inspection results. HPCI / Reactor Core Isolation Cooling (RCIC) moisture content limits, action levels and recovery actions were revised to reflect the new recommended levels. The HPCI Auxiliary Lube Oil Pump is now placed in service on a biweekly basis to recoat the EGR internals and provide real-time validation of anticipated governor valve response.

NARRATIVE

I. Plant Operating Conditions before the Event

At the time of discovery, Browns Ferry Nuclear Plant (BFN) Unit 1 was in Mode 1 at approximately 97 percent power.

II. Description of Event

A. Event Summary

On July 12, 2022, when assistant unit operators (AUOs) locally started the high pressure coolant injection (HPCI) [BJ] auxiliary oil pump [P], HPCI control valve 1-FCV-073-0019 (governor valve) [FCV] did not open as expected during the prerequisite startup checks for BFN, Unit 1 HPCI Flowrate surveillance test 1-SR-3.5.1.7. Subsequent troubleshooting identified that with the turbine at rest, and the auxiliary oil pump running, the electro-governor remote hydraulic actuator (EGR) [SM] did not immediately supply the hydraulic oil needed to open the governor valve. Operations personnel declared HPCI inoperable and entered Technical Specification (TS) 3.5.1, "Emergency Core Cooling System (ECCS) - Operating," Condition C and verified that reactor core isolation cooling (RCIC) [BN] was operable as required.

Troubleshooting confirmed that the EGR had failed due to mechanical binding as a result of accumulated corrosion product around the pilot plunger which reduced critical clearances needed for component operation. The resulting inoperability of the single-train HPCI system would have prevented HPCI from fulfilling its safety function that would be needed to mitigate the consequences of an accident. HPCI was returned to standby readiness and was declared operable on July 15, 2022, at 0915 CDT, thus exiting TS 3.5.1 Condition C.

The Tennessee Valley Authority (TVA) is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(v)(D), as any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

NARRATIVE

C. Dates and approximate times of occurrences

DATE AND APPROXIMATE TIMES	OCCURRENCE
January 5, 2000	BFN changed the frequency of periodic HPCI Auxiliary Lube Oil Pump runs from monthly to quarterly. The technical justification focused on Cold Quick Start and did not mention impacts to governor control system components, such as the EGR and Remote Servo.
October 26, 2006	0-TI-230L," Lubrication Oil Analysis & Monitoring Program," Revision 000 is issued. This establishes that the HPCI lube oil moisture content shall be less than 0.5 percent.
May 31, 2007	BFN, Unit 1 Restart
March 29, 2013	Work Order (WO) 114560684 was written to reroute the BFN, Unit 1 HPCI Gland Seal Leakoff lines downward.
February 15, 2015	0-TI-230L, "Lubrication Oil Analysis & Monitoring Program," Revision 009 is issued. This required installing a Vacudyne filter whenever lube oil moisture levels exceeded 0.2 percent.
October 29, 2018	The BFN, Unit 1 HPCI EGR Actuator and Mechanical Drive were visually inspected, during BFN, Unit 1, Refueling Outage 12 (U1R12). No adverse findings were noted at the time, despite direct photographic evidence of the EGR displaying indications of rust and/or moisture.
December 3, 2021	Electric Power Research Institute (EPRI) published 2021 Lube Notes, Note 3, "Effects of Moisture in Terry Turbine Lube Oil and Governor Systems", which provided new lube oil limit recommendations to preclude rust and other corrosion product development in HPCI and RCIC governor control system components. EPRI documented a discussion between the original equipment manufacturer and determined that a 0.5 percent lube oil moisture content was an operating limit for running turbines, where system operation mixes the oil. This led to recommending a new 0.05 percent (500 ppm) moisture warning limit, and that the 0.5 percent (5000 ppm) limit only applies to operating units.
April 13, 2022 1122 CDT	Last known successful functional operation of the BFN, Unit 1, HPCI Governor Valve EGR, which occurred during a flowrate surveillance.
July 12, 2022, 0913 CDT	BFN, Unit 1, HPCI is declared inoperable to support a routine flow rate surveillance test.

NARRATIVE

DATE AND APPROXIMATE TIMES	OCCURRENCE
July 12, 2022, 0917 CDT	AUOs started the HPCI aux oil pump locally, as part of a routine flow rate surveillance test. AUOs verified that the stop valve opened, while the governor valve unexpectedly remained closed and did not move.
July 12, 2022, 0954 CDT	The governor valve opens because of troubleshooting activities, revealing that the EGR was non-functional.
July 15, 2022, 0915 CDT	BFN, Unit 1, HPCI is declared operable following the completion of corrective maintenance and flow rate testing.
July 25, 2022	BFN receives a copy of the EPRI Lube Oil Notes which contain the revised recommended lube oil moisture content limits.
November 17, 2022	The BFN, Unit 1 HPCI Gland Seal Leakoff Lines were rerouted downward.
January 7, 2023	Rust on the BFN, Unit 1 HPCI EGR was retroactively identified based on the photographs from U1R12.

D. Manufacturer and model number of each component that failed during the event

The failed component was a Terry Steam Turbine Company turbine governor valve, Model CCS.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

Failure was discovered when the HPCI auxiliary oil pump governor valve did not open as expected during the prerequisite startup checks for a routine surveillance test.

G. The failure mode, mechanism, and effect of each failed component

The U1 HPCI EGR actuator failed due to rust accumulation on the pilot valve and inner bore, reducing the required clearances needed for the EGR to actuate when demanded.

H. Operator actions

There were no operator actions associated with this event.

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I. Automatically and manually initiated safety system responses

There were no automatic or manual safety system responses associated with this event.

III. Cause of the event

A. Cause of each component or system failure or personnel error

The procedural guidance in MPI-0-073-TRB001, "High Pressure Coolant Injection (HPCI) Turbine Preventive Maintenance", was less than adequate to ensure that visual inspection tasks identify adverse conditions prior to failure.

Additionally, the maintenance strategy was not aligned with industry best practices and was insufficient for making risk-based decisions related to the reliability of the U1 HPCI EGR.

B. Cause(s) and circumstances for each human performance related root cause

No human performance related root causes were identified.

IV. Analysis of the event

The HPCI system is provided to assure that the reactor is adequately cooled to limit fuel cladding temperature in the event of a small break in the nuclear steam supply system and loss of coolant which does not result in rapid depressurization of the reactor vessel. The HPCI system permits the nuclear plant to be shut down, while maintaining sufficient reactor vessel water inventory until the reactor vessel is depressurized. The HPCI system continues to operate until the reactor vessel pressure is below the pressure at which low pressure coolant injection (LPCI) [BO] operation or core spray system [BM] operation maintains core cooling. Due to the HPCI system's inoperability, it would have been unable to perform its safety function.

Since the BFN, Unit 1 Restart, BFN has maintained by process and procedure HPCI lube oil moisture limits which were more conservative than industry recommendations. BFN also took conservative actions as-needed when evaluating the risks of circulating moisture-laden lube oil through the EGR internals.

After the BFN, Unit 1 Restart, HPCI system monitoring and trending revealed differences between BFN Units which challenged the maintenance of lube oil moisture content levels needed to preclude corrosion. Post-2010, there was widespread industry OE regarding a poorly designed flexible wedge gate valve used for the HPCI Steam Admission Valve (SAV). This SAV design permitted 1000 psig steam leaks from the steam inlet into the turbine casing, where it would condense in the lube oil reservoir. This known moisture incursion source increased the difficulty of maintaining low-moisture oil and optimizing the longevity of EGR actuators. In 2012, BFN

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implemented the Sentinel Gate valve on the BFN, Unit 1 HPCI to mitigate the risk associated with steam leaking past the SAV. Similar modifications were performed on BFN, Units 2 and 3 in 2013 and 2014 respectively. However, this upgrade did not allow the BFN, Unit 1 HPCI lube oil moisture content to be maintained as low as that of the BFN, Unit 2 or Unit 3 HPCI.

There were no adverse events or trends regarding the BFN, Unit 1 HPCI lube oil moisture content between April 2013 and July 2018. In 2015, the HPCI lube oil performance monitoring criteria for HPCI was revised to require dewatering upon reaching a 0.2 percent moisture content, a limit which had been observed since 2008. Lube oil moisture spikes were conservatively identified, and actions were taken to ensure the newly-proceduralized moisture content threshold was not violated. At the time, a 0.2 percent moisture content was well below the EPRI-recommended value of 0.5 percent.

In 2018, an engineer performing a visual inspection during U1R12 inspected the EGR, took photos, and signed off the inspection stating that no adverse conditions were identified. These photos were archived but not reviewed again until 2023. The condition observed in 2018 revealed a degraded condition which became self-revealing when the EGR failed to actuate on July 12, 2022.

In 2020, the Unit 1 HPCI Routine PM frequency was reduced from 2 years to 4 years to support the site’s move to a Divisional Outage approach. The purpose of a Divisional Outage approach was to optimize the site’s maintenance strategy. The technical justification for the HPCI Routine PM relied on a review of previous PM performances which indicated no adverse findings were identified. This PM frequency change unknowingly exposed BFN to additional risk, as the 2018 visual examination failed to identify a condition adverse to quality with latent operability risks. This PM performance review was cited again in 2021, when BFN evaluated the adoption of a new PM for periodic auxiliary lube oil pump (ALOP) runs to exercise the governor system and rewet the EGR internals. This PM was not implemented since there were no adverse moisture trends and BFN was performing to criteria more conservative than what EPRI recommended at the time. In December 2021, EPRI published a new moisture content limit recommendation of 0.05 percent (500 ppm) to minimize the development of EGR internal corrosion. BFN did not recognize this recommendation until late July 2022.

On July 12, 2022, during performance of the quarterly flowrate surveillance, the governor valve was observed to be closed, and not in the expected open position, during the prestart up trip test sequence. The surveillance was suspended, and troubleshooting confirmed that the EGR had failed due to mechanical binding from an accumulation of corrosion products around the pilot plunger, which reduced critical clearances needed for component operation. During the initial engineering evaluation for this event, BFN identified inadequate industry guidance as a factor that led to the event. During this evaluation, the photos from the U1R12 visual inspection which displayed degraded EGR internals remained undiscovered, because such photographs were not a required System Monitoring activity at the time. These photographs were located and reviewed

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on January 7, 2023. After validating that these images were applicable to a previously documented satisfactory visual inspection, CR 1827759 was generated to document the retroactive identification of the BFN, Unit 1 HPCI EGR internal corrosion.

The governor valve opened on July 12, 2022, at 0954 CDT. This time is being taken as the time of discovery because that is when cognizant individuals were notified that governor valve would have been unable to open upon receiving a valid demand signal.

V. Assessment of Safety Consequences

This event resulted in inoperability and unavailability of the single train of the BFN, Unit 1, HPCI system resulting in the inability of the HPCI system to perform its safety functions to mitigate the consequences of an accident. Additionally, On June 1, 2022, RCIC was inoperable between 0110 CDT and 0140 CDT, to support surveillance testing. In the event of an emergency, all Automatic Depressurization Systems (ADS) were available during this event to facilitate core cooling by low pressure Emergency Core Cooling Systems (ECCS), and at all other times, the RCIC system remained operable. A Probabilistic Risk Assessment (PRA) of the event determined that the incremental core damage probability (ICCDP) delta and incremental conditional large early release probability (ICLERP) delta resulted in a Core Damage Frequency (CDF) of 5.65E-06 and Large Effluent Release Frequency (LERF) of 2.41E-07.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

On June 1, 2022, RCIC was inoperable between 0110 CDT and 0140 CDT, to support surveillance testing. Otherwise, RCIC and all other ECCS and ADS systems remained operable for the duration of the event.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shutdown.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

It is assumed that the Unit 1 HPCI was unavailable for an exposure time of T/2, plus the repair time. The last successful functional operation of the component was on April 13, 2022, and the time of discovery was July 12, 2022. Therefore, the T/2 date is May 28, 2022, meaning that HPCI is considered inoperable for 45 days prior to discovery. The Unit 1 HPCI Turbine Control Valve

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was declared operable following post-maintenance testing approximately 3 days later, on July 15, 2022. The BFN, Unit 1, HPCI system was inoperable for approximately 48 days.

VI. Corrective Actions

Corrective Actions are being managed by the TVA's corrective action program under CRs 1789217 1827759, and 1829973.

A. Immediate Corrective Actions

The EGR and remote servo were promptly replaced.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

As corrective actions to prevent recurrence, BFN has:

- Revised the section of MPI-0-073-TRB001, "High Pressure Coolant Injection (HPCI) Turbine Preventive Maintenance" applicable to the EGR visual inspection. The revised procedure now includes enhanced acceptance criteria (both wording and images) to remove the task performer from knowledge-based performance and ensure that the task will be performed in rule-based performance during future inspections. The new inspection instructions will require CR initiation and photographing the inspection results.
- Revised 0-TI-230L, "Lubrication Oil Analysis & Monitoring Program", to include HPCI / RCIC moisture content limits, action levels and recovery actions to reflect the new recommended moisture limits and CAPR requirements.
- Established a new Preventive Maintenance Activity to place the HPCI Auxiliary Lube Oil Pump in service on a biweekly basis to recoat the EGR internals and provide a real-time validation of anticipated governor valve response.

VII. Previous Similar Events at the Same Site

A search of LERs from BFN, Units 1, 2, and 3 over the last five years identified no similar events.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.